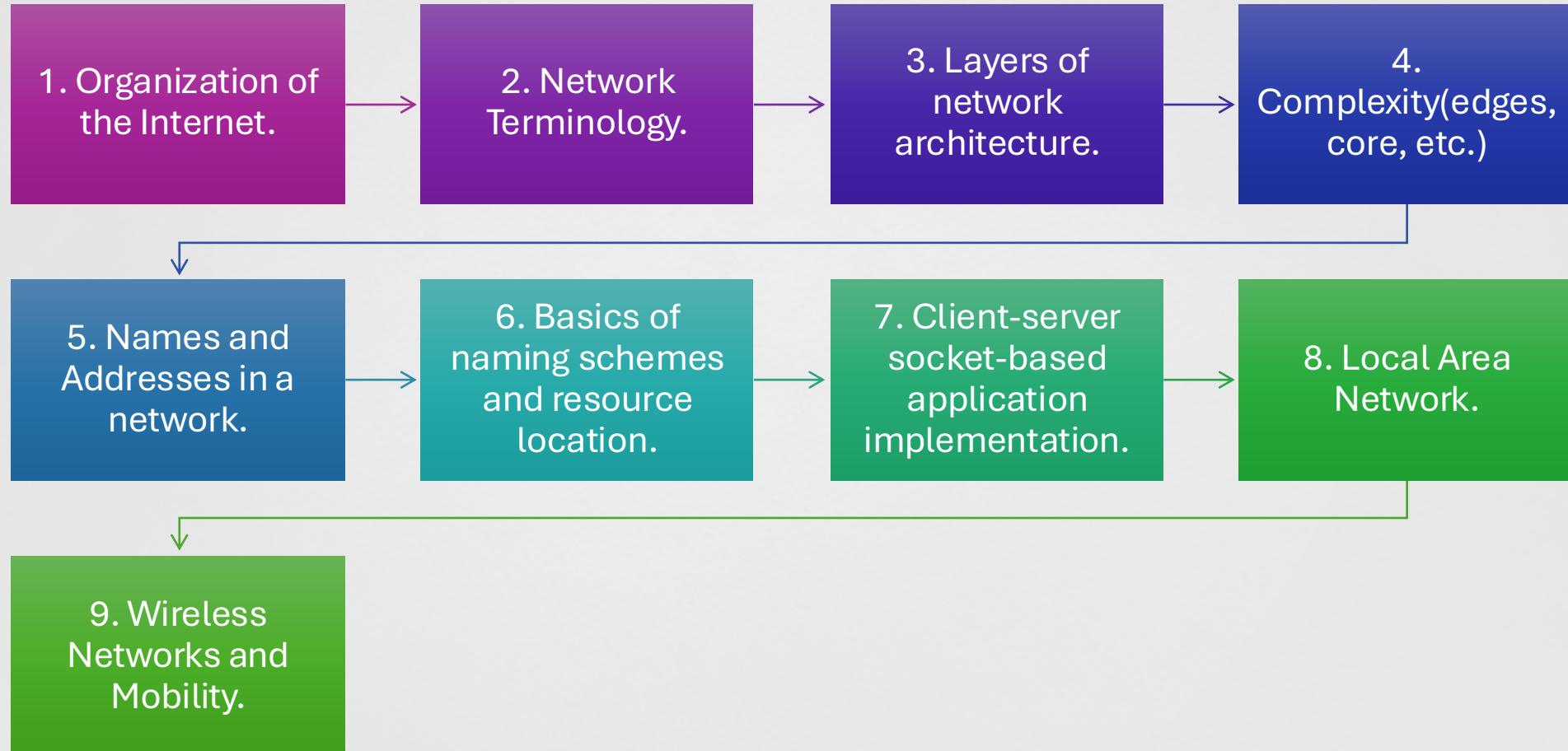




Introduction to Networking and Communication

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Topics to be covered



What do we mean
by the
organization of
the Internet?

- When we talk about the **organization of the Internet**, we refer to how it is structured, managed, and coordinated to ensure it functions smoothly across the globe. It's not a single entity but a complex system of networks, protocols, and governing bodies that collaborate to provide global connectivity. Let's break it down clearly.

Global network of networks.

- The internet is not owned by a single organization
- It's a network of networks.
- (What do we mean by network of networks?) \
- We mean that the Internet itself is not a single network, but a collection of many smaller networks that are connected.
- Individual networks –like those of **Internet Service Providers (ISPs)**, universities, or companies –are interconnected to exchange data.
- These connections form a hierarchical structure:
- Local networks → , Regional networks → , Global backbone networks

-

Global
network of
networks.



The internet infrastructure is simply a network of networks. AERIALPERSPECTIVE IMAGES/GETTY IMAGES

Hierarchical Diagram of Networks

→ International Backbone (Tier 1 ISPs)

These are global Internet providers that form the Internet's high-speed backbone and interconnect major networks worldwide.

→ National / Regional ISPs (Tier 2 ISPs)

They purchase connectivity from Tier 1 ISPs and distribute Internet access across specific countries or large regions.

→ Local ISPs (Tier 3 ISPs)

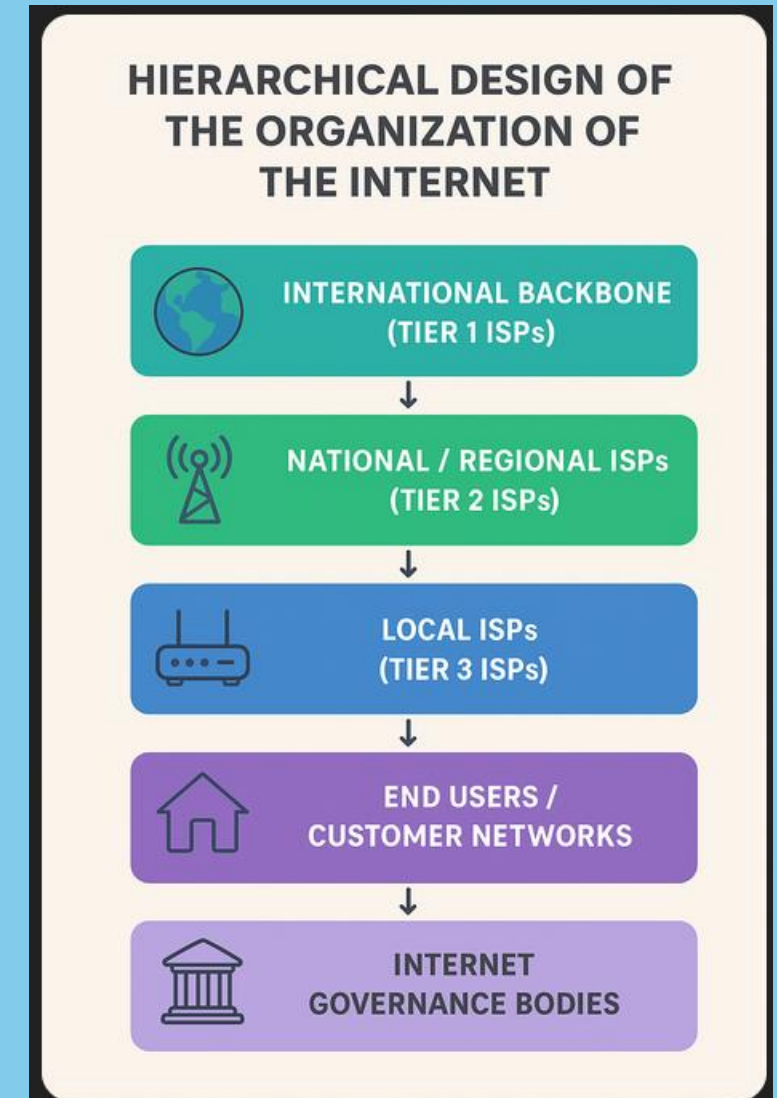
They provide direct Internet connections to homes, offices, and local users through broadband, fiber, or wireless access.

→ End Users / Customer Networks

These are individuals, organizations, and institutions that use Internet services through their local network connections.

→ Internet Governance Bodies

These organizations (like ICANN, IETF) manage Internet standards, IP addresses, and domain names to ensure global coordination.



2. Network Terminology.

- **Network Terminology** refers to the **common technical words and concepts** used in computer networking to describe how data is transmitted, connected, and managed between devices.
- These terms form the **basic language of networking**, helping us understand how networks (like the Internet or LANs) operate.

- **These are some common network terminologies >>>>>>>>>>**

Term	Meaning / Definition
Node	Any device connected to a network (like a computer, phone, or printer).
IP Address	A unique number that identifies a device on a network (e.g., 192.168.1.1).
Router	A device that connects different networks and directs data between them.
Switch	Connects multiple devices in a local network and sends data to the correct destination.
Protocol	Rules that define how data is transmitted between devices (e.g., TCP/IP).
Packet	A small piece of data sent across the network.
Bandwidth	The amount of data that can be transferred per second.
Latency	The time delay in data transmission.
DNS	Converts website names (like www.google.com) into IP addresses.
LAN (Local Area Network)	A network covering a small area, such as an office or home.
WAN (Wide Area Network)	A network that covers a large area, connecting multiple LANs (e.g., the Internet).

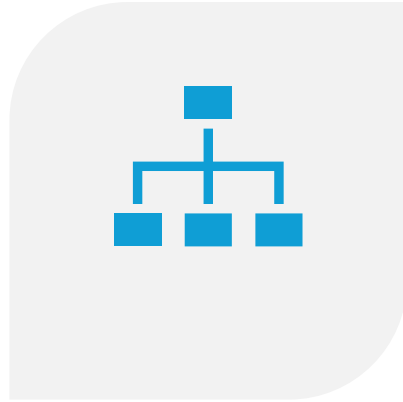
Network Terminology (With Advanced Insight)

- Network terminology includes the core concepts and components that describe how computers and devices communicate across local and global networks. Understanding these terms helps you see how data travels, how devices connect, and how networks stay secure and efficient.
- Topic# Network Terminology.

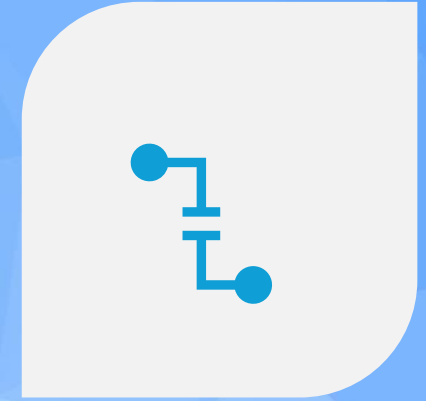
3. Layers of network architecture.



THE LAYERS OF NETWORK ARCHITECTURE REFER TO THE STRUCTURED DESIGN OF HOW DATA TRAVELS FROM ONE DEVICE TO ANOTHER IN A NETWORK.



INSTEAD OF DOING EVERYTHING AT ONCE, NETWORKING IS DIVIDED INTO LAYERS, WHERE EACH LAYER HAS A SPECIFIC ROLE — FOR EXAMPLE, ONE LAYER HANDLES DATA TRANSMISSION, ANOTHER DEALS WITH ADDRESSING, AND ANOTHER ENSURES THE MESSAGE IS DISPLAYED CORRECTLY TO THE USER.



THIS LAYERED APPROACH MAKES NETWORKS EASIER TO DESIGN, UNDERSTAND, TROUBLESHOOT, AND UPGRADE.

The First common layer model-(OSI model)

- 1 The **OSI Model** provides a **standard framework** for understanding and designing network communication systems.
- 2 It helps **identify and troubleshoot** network issues by isolating problems to specific layers.
- 3 It enables **interoperability between different hardware and software** through well-defined communication protocols.
- 4 It supports **network security** by applying encryption, authentication, and access control at specific layers.
- 5 It enhances **application development** by allowing software to communicate efficiently across networks.

Topic# Layer of network architecture.

HIERARCHICAL DESIGN OF THE OSI MODEL



Application Layer

Provides network services directly to user applications like browsers and email
Ex. HTTP, SMTP, FTP, DNS



Presentation Layer

Translates, encrypts, or compresses data for the Application layer
Ex. SSL/TLS, JPEG, MPEG



Session Layer

Manages, establishes, and terminates sessions between applications
Ex. NetBIOS, RPC



Transport Layer

Provides reliable or unreliable delivery and flow control
Ex. TCP, UDP



Data Link Layer

Responsible for node-to-node data transfer and error detection
Ex. IP, ICMP, OSPF, Switches



Physical Layer

Deals with transmission of raw bits over physical media (cables, radio waves)
Ex. Cables, Hubs, Fiber optics

The second common layer model-(TCP/IP Model

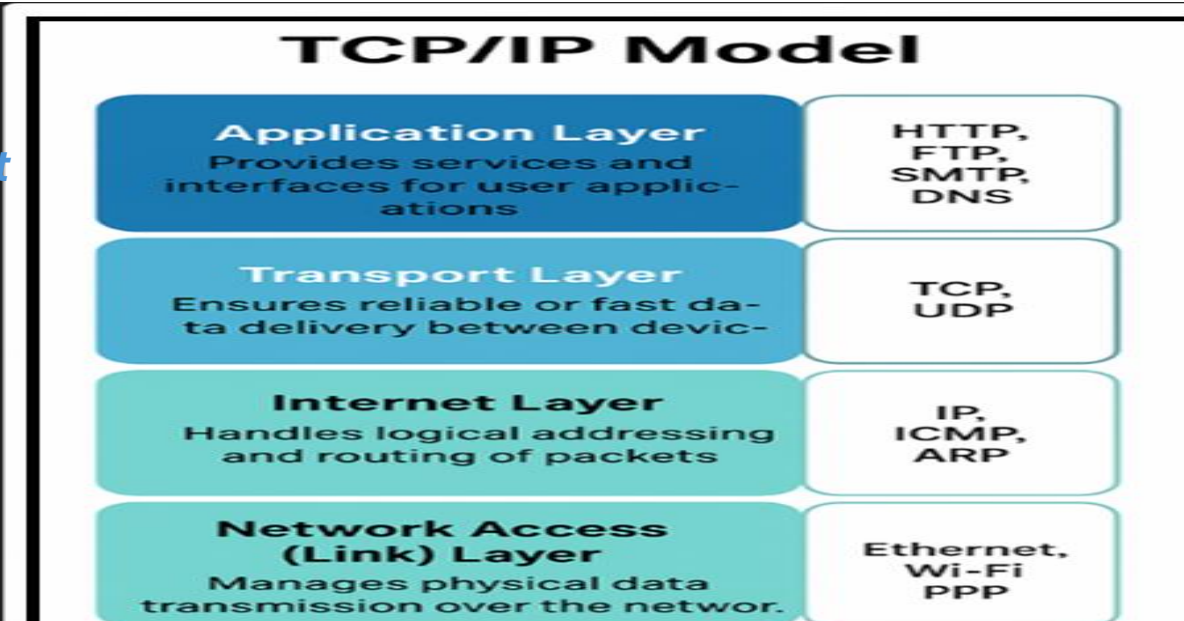
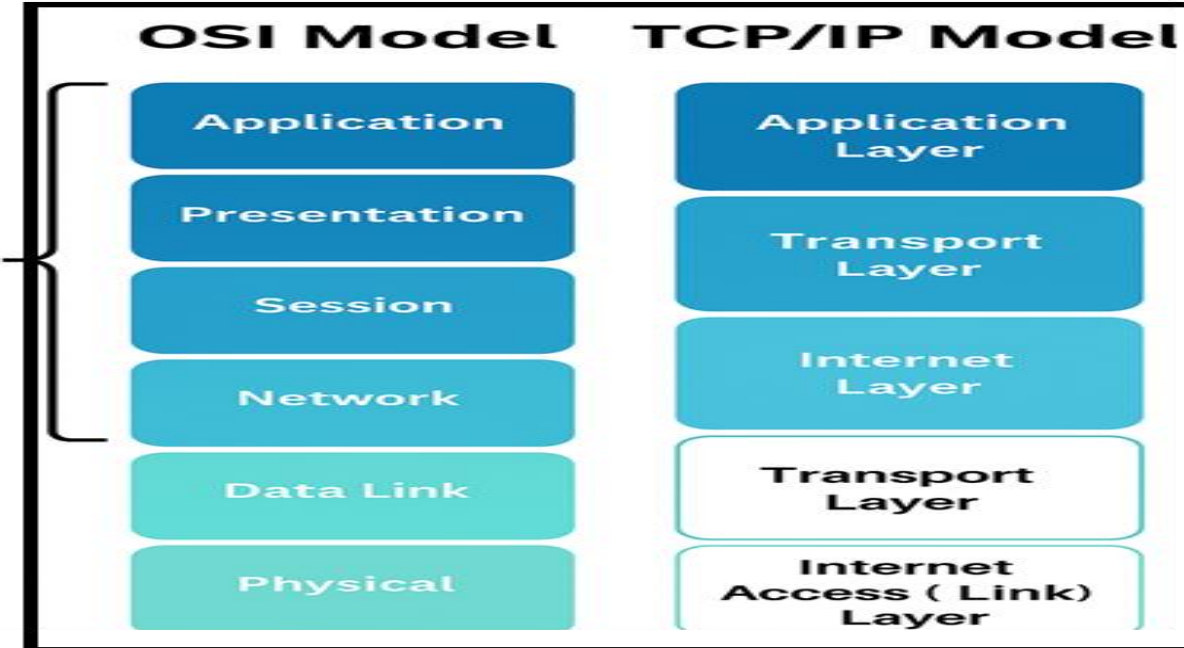
Common of Both Layers

Benefits of the TCP Layer (Transport Layer in TCP/IP Model)

- 1 **Reliable Data Transmission:** Ensures that data packets are delivered accurately and in the correct order using acknowledgment and retransmission.
- 2 **Error Detection and Flow Control:** Manages congestion and prevents data loss by controlling the rate of transmission.
- 3 **End-to-End Communication:** Provides logical connections between sender and receiver applications, making communication seamless and secure.

The TCP/IP layer model was created to provide a practical, simpler, and more efficient framework for real-world Internet communication, whereas the OSI model was mainly theoretical and conceptual.

In short, OSI explains how networks should work; TCP/IP shows how they actually work.



4. Complexity (Edges, Core, etc)



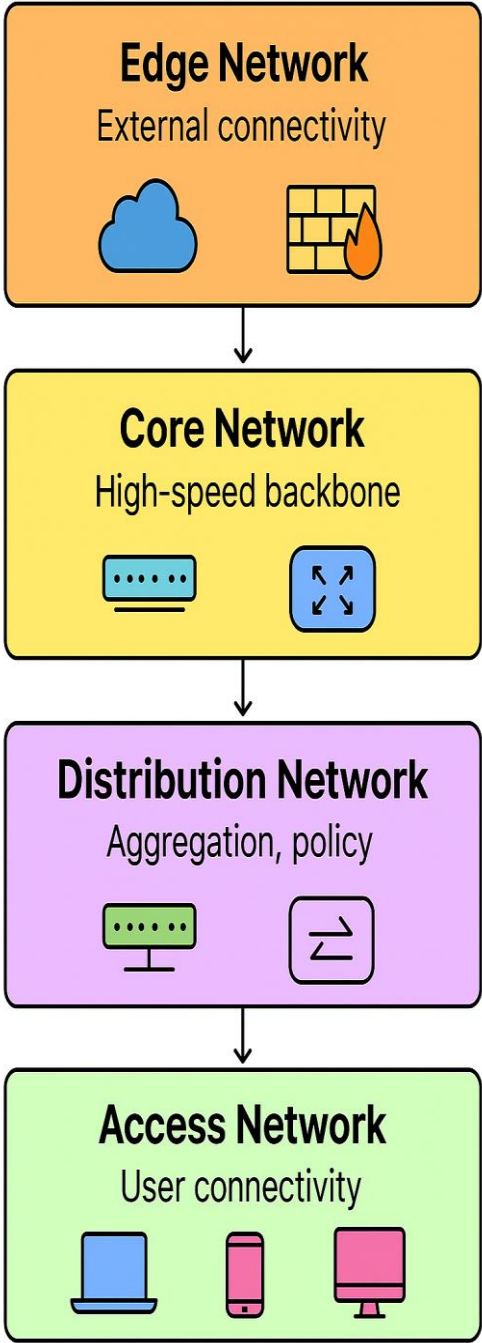
In computer networking, **network complexity** refers to how different parts of the Internet or a large network are organized and connected to handle data flow efficiently.



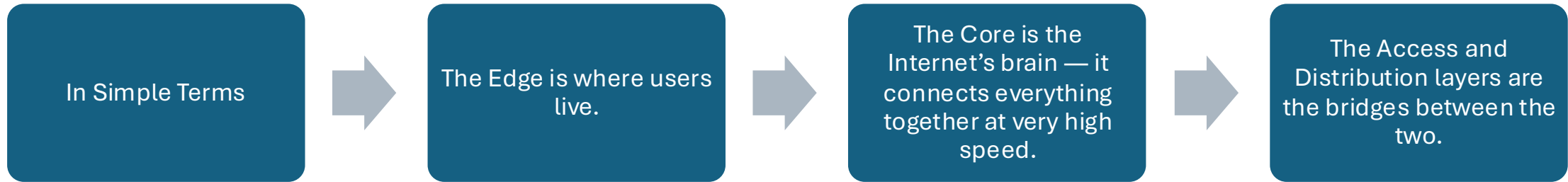
The Internet isn't one flat structure — it's **hierarchical**, made up of layers of interconnected networks.

Here's how it's usually divided:

Component	Description	Main Function	Example
Edge Network	The outermost part of the Internet — where users and devices connect.	Provides access to end users (computers, phones, IoT devices).	Home Wi-Fi, office LAN, mobile networks.
Access Network	Connects users to Internet Service Providers (ISPs).	Provides connectivity between users and ISPs .	DSL, cable, fiber, or cellular network.
Core Network	The central backbone of the Internet connecting major ISPs and data centers.	Handles high-speed data transfer and routing across long distances.	Internet backbone routers, optical fiber links.
Distribution Network	Connects edge and core layers.	Manages traffic aggregation and routing between local and backbone networks.	Regional ISP networks.



Complexity(Edges, Core, etc) {Advance sight}



Why it is called “Network Complexity”

Because managing:

Multiple layers (edge, core, etc.),

Different devices (routers, switches, servers),

And vast routing protocols (like BGP, OSPF)

makes large networks complex — requiring careful design to ensure speed, reliability, and scalability.

5. Name and addresses in the network.

- In computer networking, names and addresses are essential concepts used to identify and locate devices, users, and services. Every system connected to a network must have a unique identity so that data can be correctly sent, received, and managed.
- While names are designed for human understanding, addresses are used by machines for accurate data routing and delivery. Together, they form the foundation of how communication takes place over the Internet and local networks.

Names and addresses in network (Detailed)

Every device or service on a network must have a unique identity so that data can be sent and received correctly.

Names are human-readable identifiers, e.g→, (www.google.com) used to easily recognize network resources.

Addresses are machine-readable identifiers, e.g→, (192.168.1.10) used by devices to locate each other for data transmission.

A mapping system such as DNS (Domain Name System) converts names into IP addresses, and ARP (Address Resolution Protocol) converts IP addresses into MAC addresses.

Together, names and addresses allow smooth communication by linking what users type to where the data actually goes.

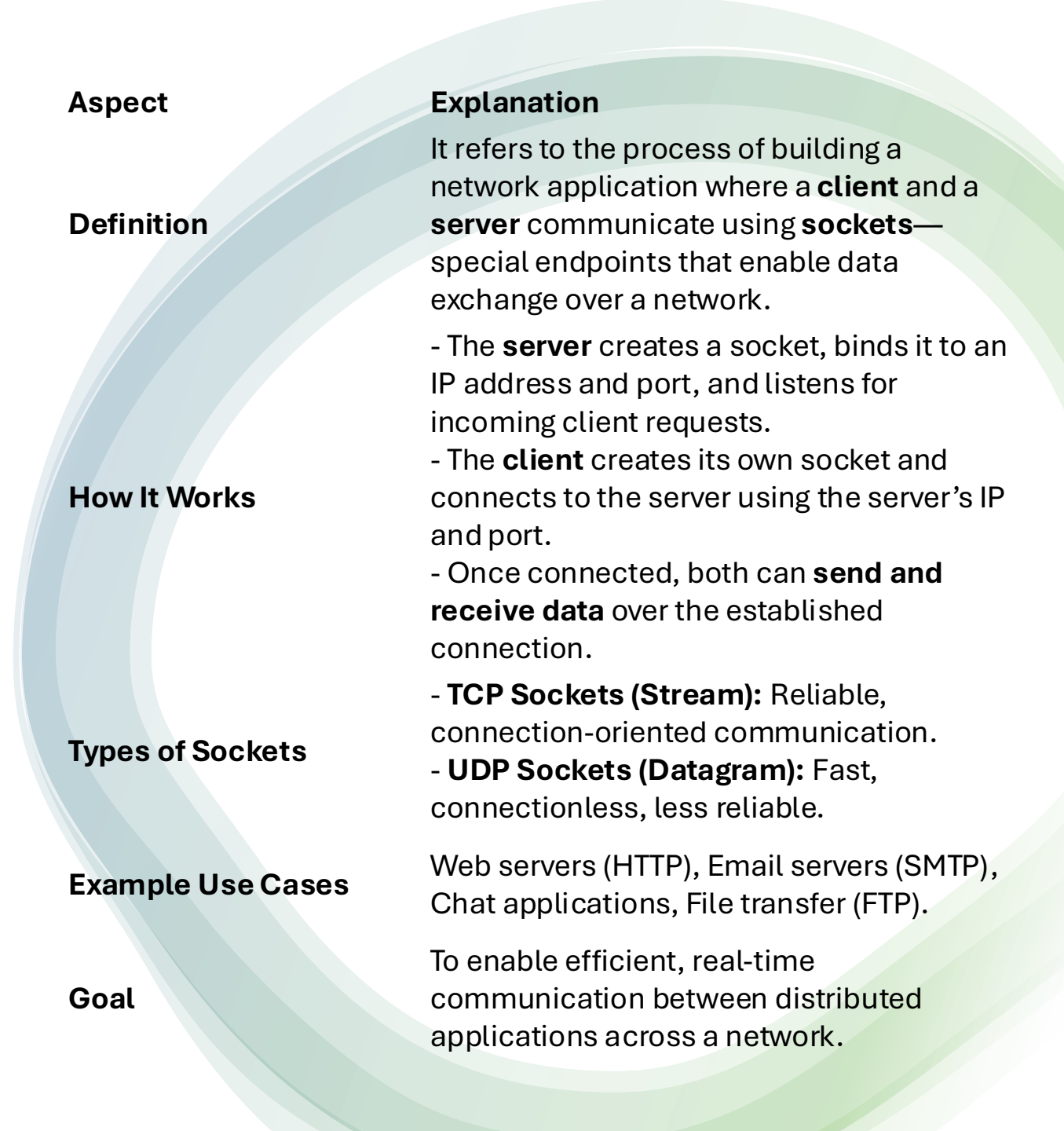
They form the basis of Internet communication, routing, and resource access, ensuring that every packet reaches the correct destination efficiently.

6. Basics of naming schemes and resource location.

- In computer networks, naming schemes and resource location are essential for identifying and finding devices, services, or data across a network.
- A naming scheme is a structured way of assigning names to network resources—such as computers, printers, websites, or files—so that they can be easily recognized and accessed by users or applications. For example, a website name like (*www.example.com*)
- Example:
When you type **www.google.com** the **DNS** translates that name into an **IP address (like 142.250.190.14)** so your computer can locate and connect to **Google's web server**.

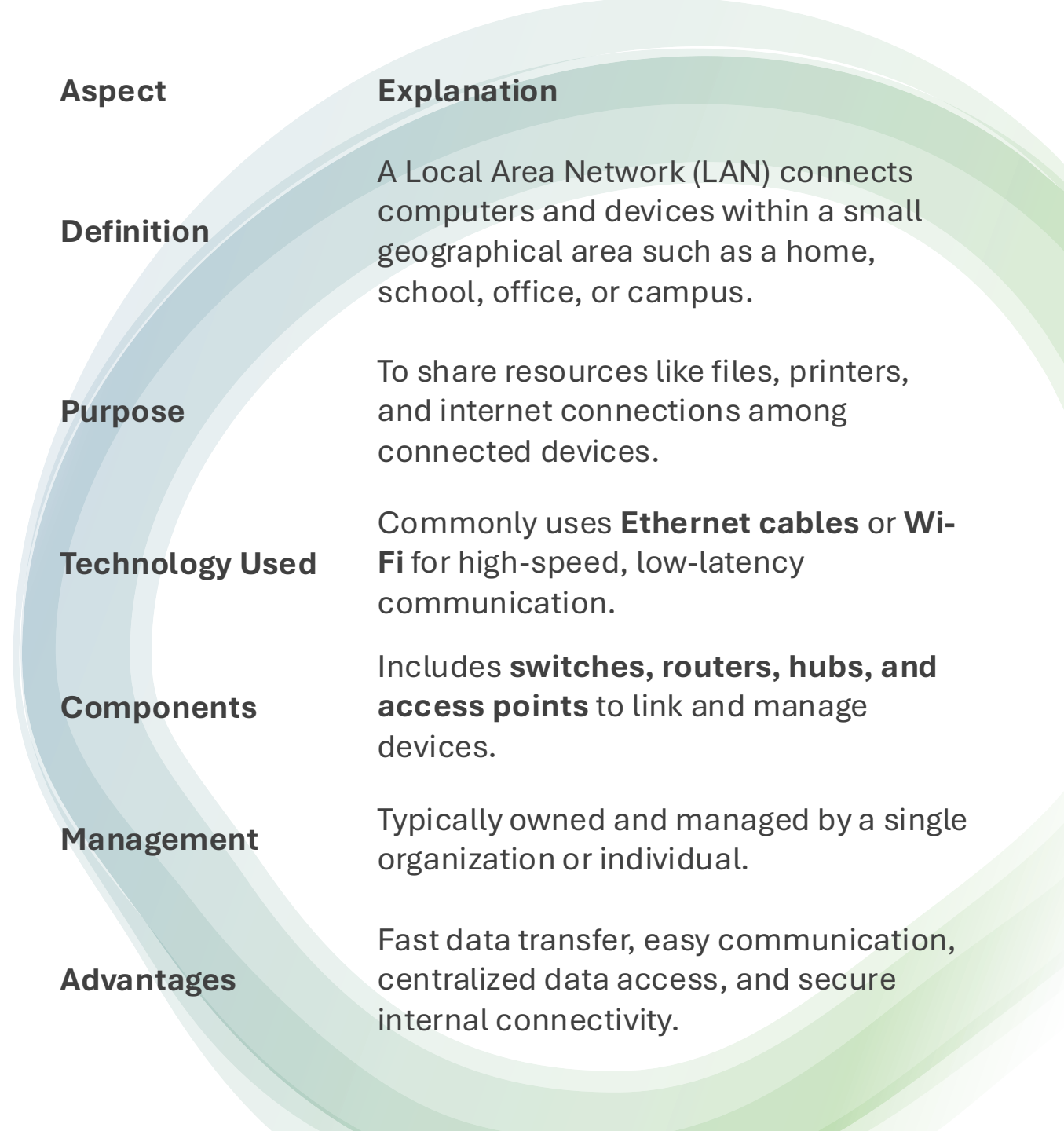
7. Client-server socket-based application implementation.

- Client-Server Socket-Based Application Implementation refers to the process of developing network applications where a client and a server communicate using sockets — software endpoints that allow data to be sent and received over a network. In this model, the server waits (listens) for client connection requests through a specific port, and once connected, both exchange data through established sockets. This mechanism enables real-time communication in applications like web browsing, chat systems, and file transfer.



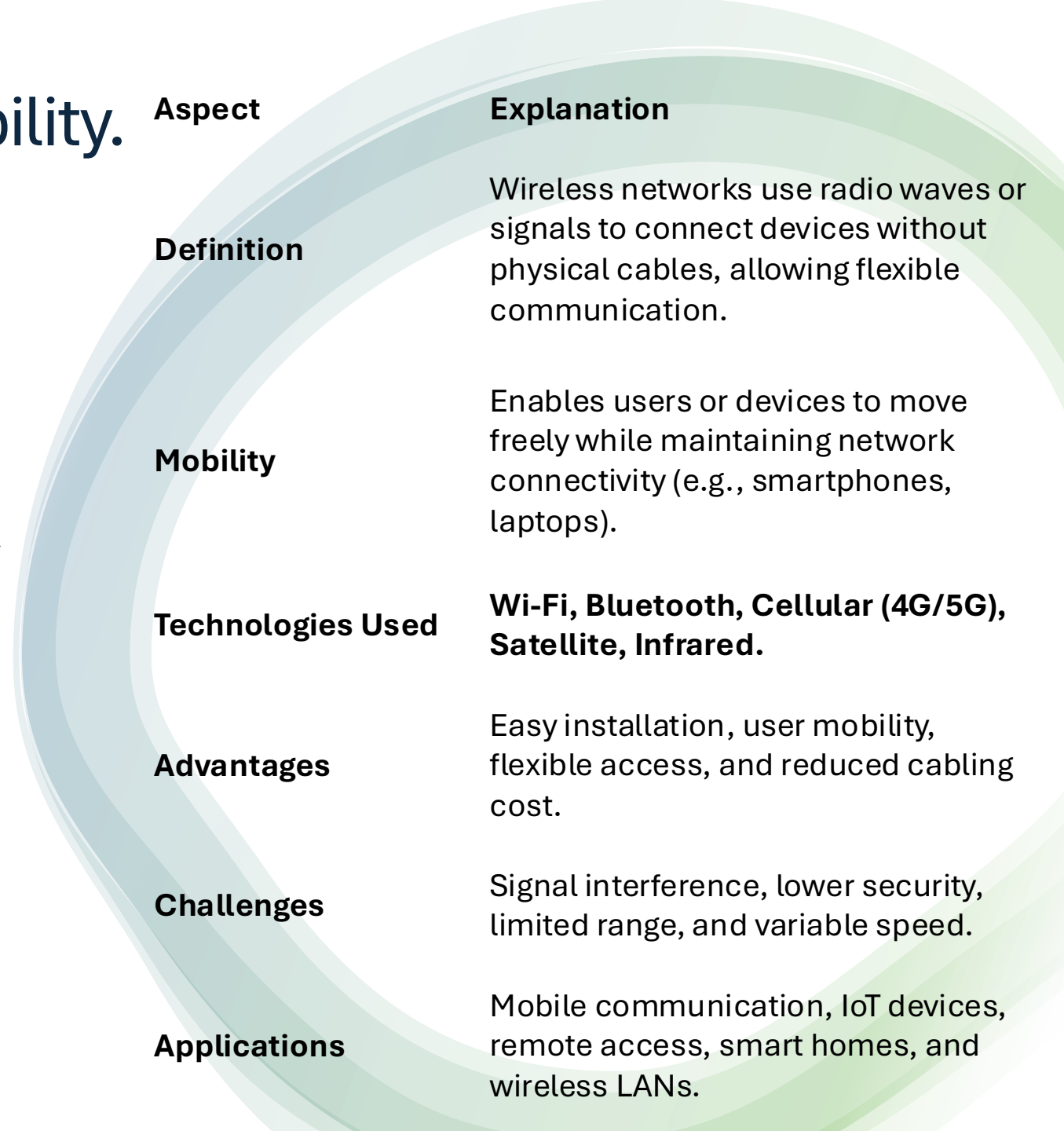
8. Local Area Network.

- A Local Area Network (LAN) is a communication network that connects computers, servers, and other devices within a limited geographical area, such as a home, office, or campus.
- It allows users to share resources like files, printers, and internet connections efficiently.
- LANs typically use Ethernet cables or Wi-Fi for high-speed data transfer and low latency. They are managed by a single organization and often include switches, routers, and access points for connectivity.
- LANs form the foundation for internal communication, collaboration, and centralized data management in modern networks.



9. Wireless Networks and Mobility.

- Wireless Networks and Mobility refer to communication systems that allow devices to connect and exchange data without using physical cables.
- Wireless networks use radio waves, infrared, or satellite signals to transmit information between mobile devices, access points, and servers. Mobility enables users to move freely while staying connected to the network, such as with smartphones, laptops, or IoT devices.
- Together, they form the basis of technologies like Wi-Fi, cellular networks (4G/5G), Bluetooth, and satellite communication, supporting seamless, flexible, and location-independent access to network resources.



QUESTIONS
PLS





THANKS TO YOU AND
WISH YOU BEST LUCK